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1. INTRODUCTION

1.1 Project Overview

Heart disease is one of the leading causes of death worldwide. Early identification of risk factors and patterns in patient data can help healthcare professionals take preventive actions and improve patient outcomes. However, analyzing large healthcare datasets manually can be difficult and time-consuming.

The **Heart Disease Analysis Project** aims to analyze heart disease related data using **Tableau and Tableau Prep Builder** to uncover important insights and patterns. The project focuses on preprocessing the dataset, transforming and cleaning the data using Tableau Prep Builder, and creating interactive dashboards in Tableau to visualize trends and relationships between different health factors such as age, cholesterol levels, blood pressure, chest pain type, and heart disease presence.

Through visual analytics, the project helps healthcare stakeholders better understand risk factors associated with heart disease. The dashboards allow users to explore patient data interactively, identify high-risk groups, and gain insights that can support data-driven healthcare decisions.

1.2 Purpose

The purpose of this project is to **analyze and visualize heart disease data to identify key health indicators and patterns associated with heart disease risk**. By using Tableau visualization tools, the project transforms raw healthcare data into meaningful and interactive insights.

The main objectives of the project include:

- Cleaning and preparing the dataset using **Tableau Prep Builder**.
- Analyzing patient health indicators such as age, cholesterol, blood pressure, and chest pain types.
- Identifying patterns and trends related to heart disease occurrence.
- Developing **interactive Tableau dashboards** to present insights clearly.
- Supporting healthcare professionals and analysts in understanding risk factors through data visualization.

The project ultimately aims to demonstrate how **data visualization tools like Tableau can help simplify complex healthcare data and improve decision-making in medical analysis.**

2. IDEATION PHASE

2.1 Problem Statement :

Create a problem statement to understand your customer's point of view. The Customer Problem Statement template helps you focus on what matters to create experiences people will love.

A well-articulated customer problem statement allows you and your team to find the ideal solution for the challenges your customers face. Throughout the process, you'll also be able to empathize with your customers, which helps you better understand how they perceive your product or service.

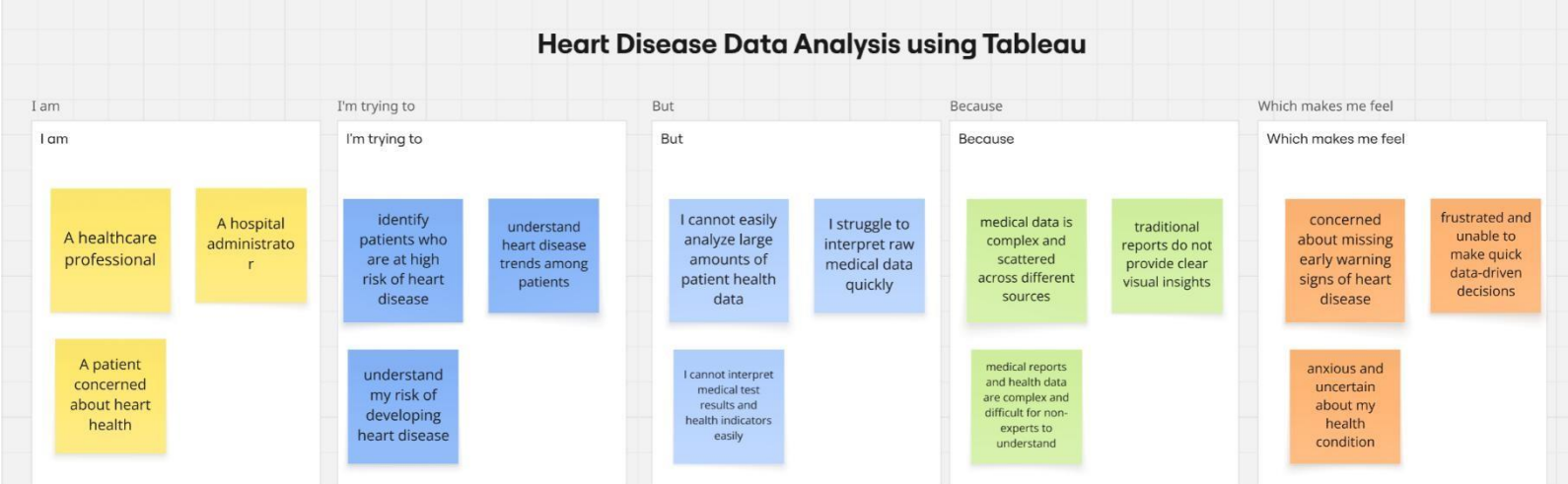


Figure 1Problem Statement Chart

Problem Statement (PS)	I am	I'm trying to	But	Because	Which makes me feel
PS-1	A healthcare professional	identify patients who are at high risk of heart disease	I cannot easily analyse large amounts of patient health data	medical data is complex and scattered across different sources	concerned about missing early warning signs of heart disease
PS-2	A hospital administrator	understand heart disease trends among patients	I struggle to interpret raw medical data quickly	traditional reports do not provide clear visual insights	frustrated and unable to make quick data-driven decisions

PS-3	A patient concerned about heart health	understand my risk of developing heart disease	I cannot interpret medical test results and health indicators easily	medical reports and health data are complex and difficult for non-experts to understand	anxious and uncertain about my health condition

2.2 Empathy Map Canvas:

An empathy map is a simple, easy-to-digest visual that captures knowledge about a user's behaviours and attitudes.

It is a useful tool to helps teams better understand their users.

Creating an effective solution requires understanding the true problem and the person who is experiencing it. The exercise of creating the map helps participants consider things from the user's perspective along with his or her goals and challenges.

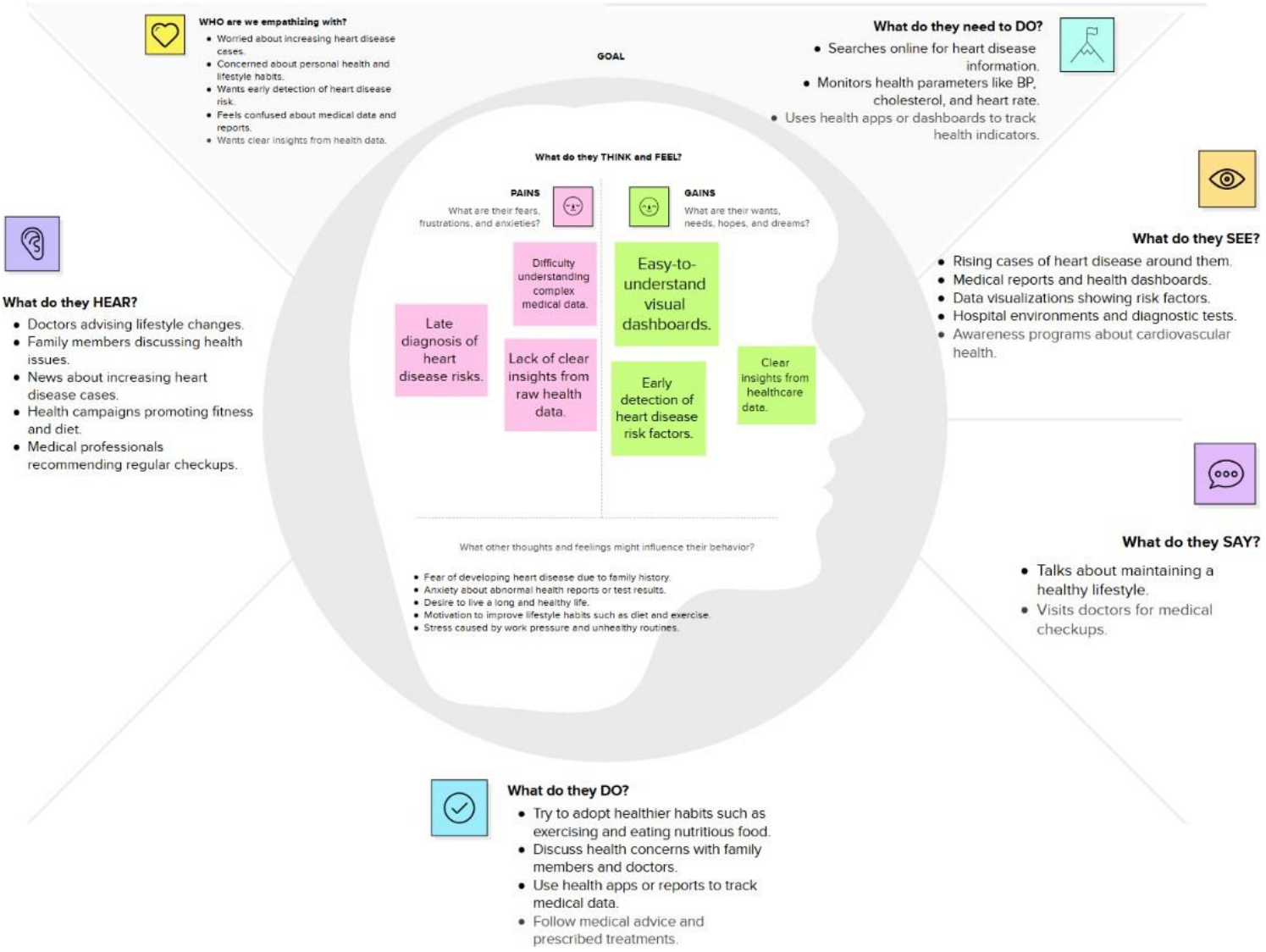


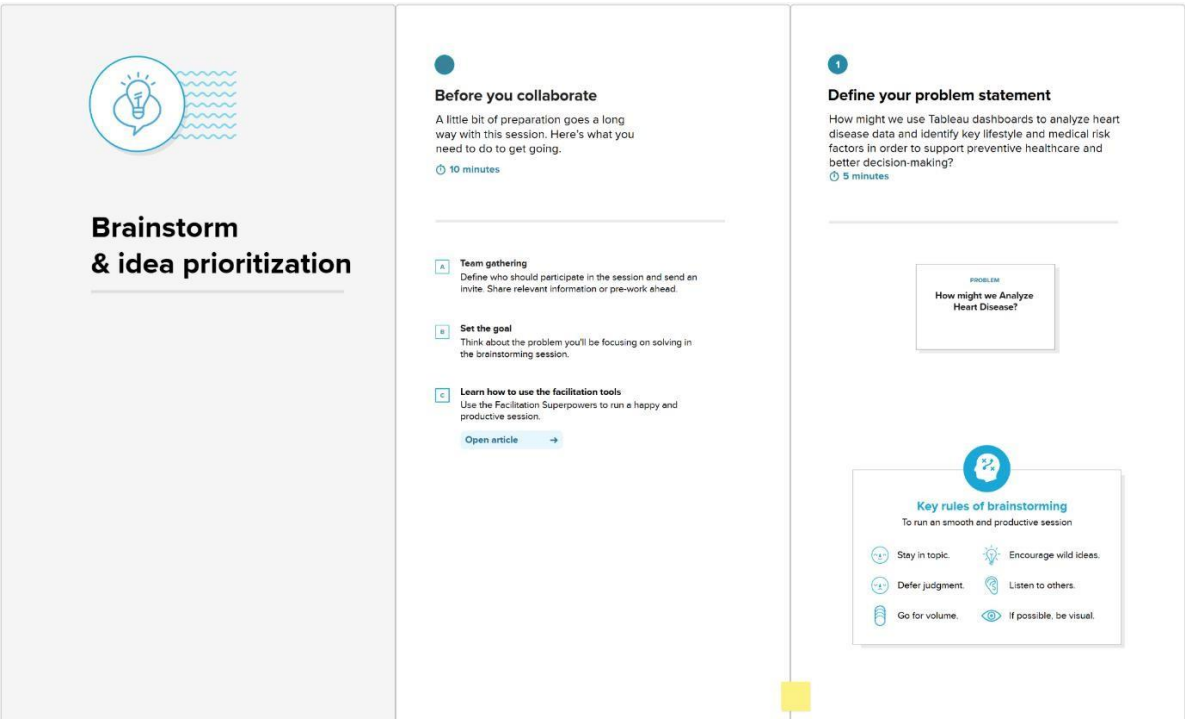
Figure 2 Emapthy Map

2.3 Brainstorm & Idea Prioritization Template:

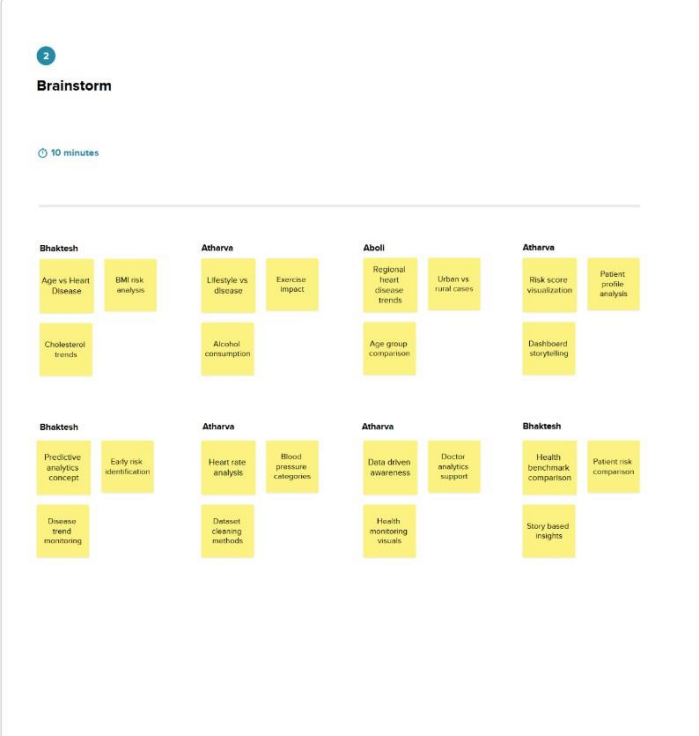
Brainstorming provides a free and open environment that encourages everyone within a team to participate in the creative thinking process that leads to problem solving. Prioritizing volume over value, out-of-the-box ideas are welcome and built upon, and all participants are encouraged to collaborate, helping each other develop a rich amount of creative solutions.

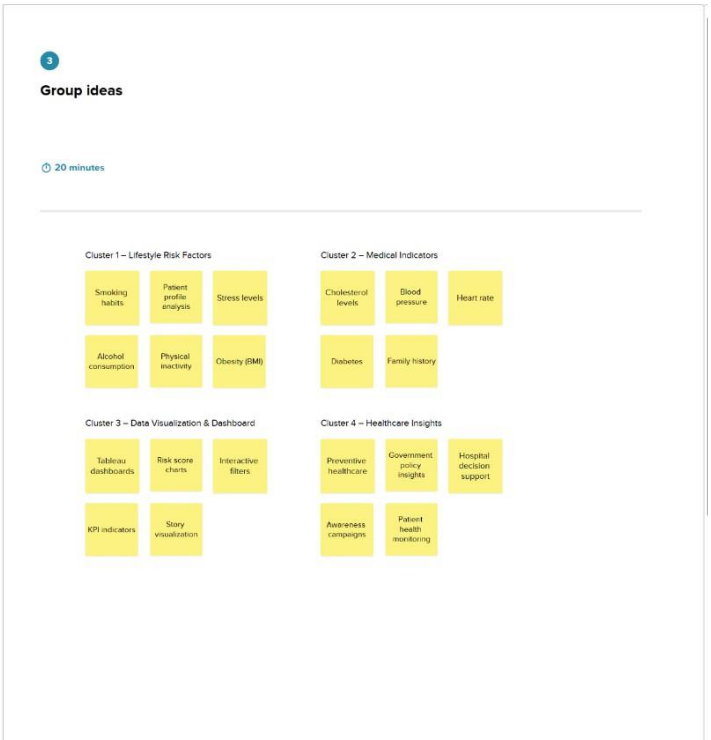
Use this template in your own brainstorming sessions so your team can unleash their imagination and start shaping concepts even if you're not sitting in the same room.

Step-1: Team Gathering, Collaboration and Select the Problem Statement

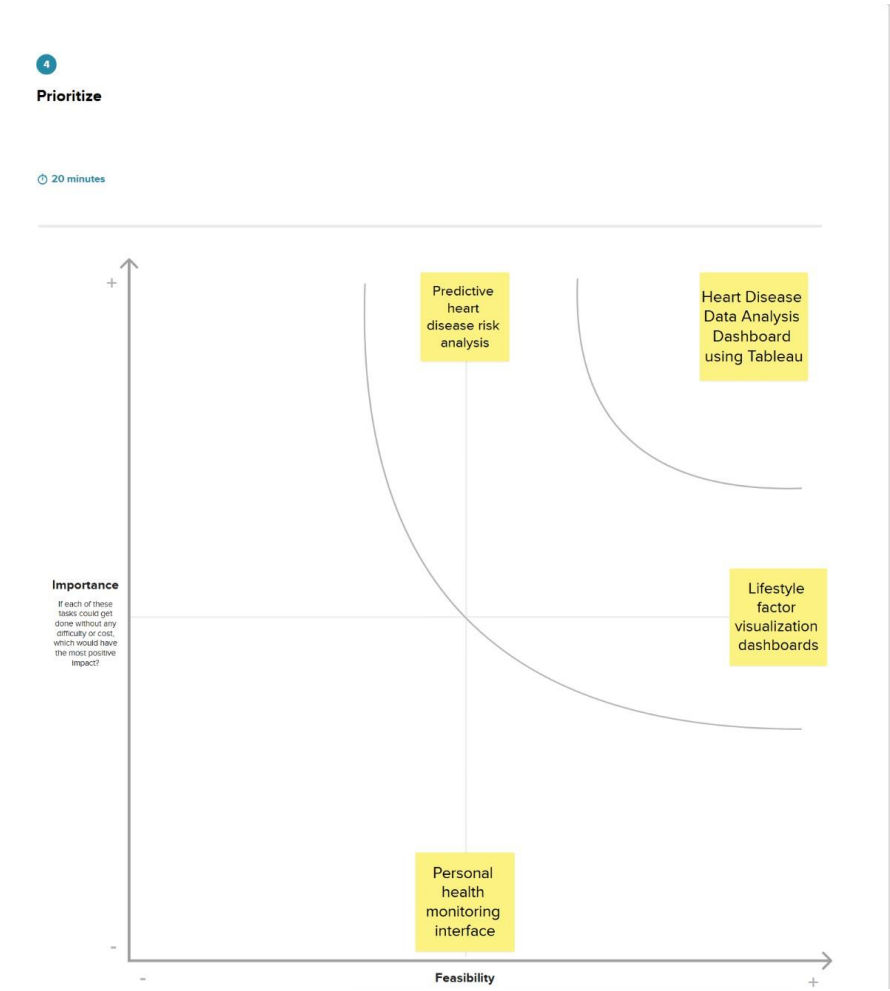


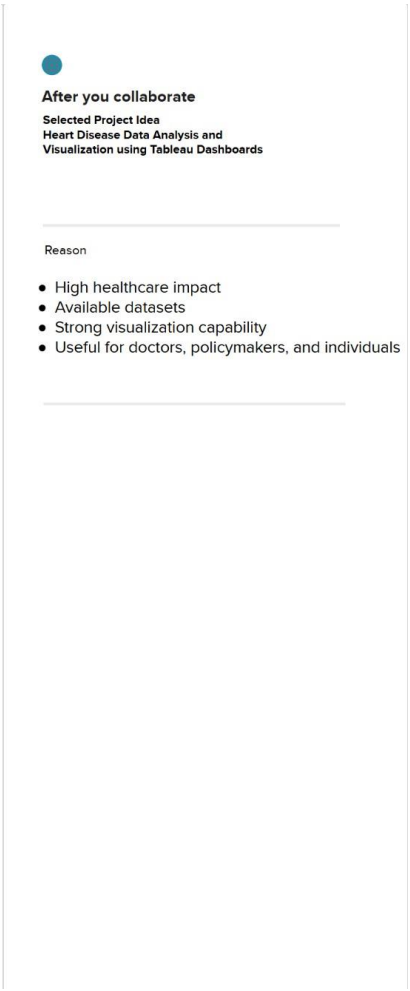
Step-2: Brainstorm, Idea Listing and Grouping





Step-3: Idea Prioritization





3. REQUIREMENT ANALYSIS

3.1 Customer Journey Mapping

User persona

Healthcare Analyst / Hospital Data Manager who analyzes patient health records using Tableau dashboards to identify disease patterns and support doctors in decision-making.

User goal

To analyze healthcare datasets efficiently and generate clear visual insights about disease risk factors, patient demographics, and treatment outcomes using Tableau dashboards.

Scenario

A healthcare analyst receives hospital patient data and uses Tableau to visualize patterns such as age distribution, cholesterol levels, blood pressure trends, and heart disease risk indicators. These insights help doctors understand disease patterns and support preventive healthcare decisions.

User backstory

The analyst works in a hospital analytics department. Previously, analyzing patient records manually in spreadsheets was time-consuming and prone to errors. The organization adopted **Tableau and Tableau Prep Builder** to clean data and create interactive dashboards for better healthcare insights.

Journey map

Stage	Data Collection	Data Preparation	Data Visualization	Insight Generation	Decision Support
User action	Gather patient data from hospital databases and medical records	Clean, filter, and structure healthcare data	Create charts, graphs, and dashboards in Tableau	Analyze trends like age, cholesterol, blood pressure, and disease risk	Share dashboards with doctors and hospital management
Pain points or questions	• Data comes from multiple sources and formats	Missing values or inconsistent records	Difficulty choosing correct visualization	Identifying meaningful patterns can be complex	Ensuring data is understandable for non-technical users
Emotions	Confused	Frustrated	Curious	Analytical	Confident
Opportunities and solutions	• Use **Tableau Prep Builder** for data cleaning and integration	Implement standardized data preparation workflow	Provide predefined healthcare dashboard templates	Use calculated fields and filters for deeper analysis	Interactive dashboards and clear visual storytelling

Emotions Across Journey

Confusion → Frustration → Curiosity → Analytical Thinking → Confidence

Opportunities

- Automated **data preparation using Tableau Prep Builder**
- Interactive **healthcare dashboards**

- Real-time **patient risk visualization**
- Better **decision support for doctors**
- Improved **data-driven healthcare management**

3.2 Solution Requirements (Functional & Non-functional)

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Data Upload	Upload heart disease dataset (CSV/Excel)
		Import patient medical records
FR-2	Data Processing	Clean and preprocess medical data
		Handle missing values and incorrect entries
FR-3	Data Analysis	Analyze risk factors such as age, cholesterol, blood pressure, etc.
		Identify patterns related to heart disease
FR-4	Data Visualization	Generate charts and dashboards using Tableau
		Display insights such as risk distribution, patient statistics, and trends

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The system should provide an easy-to-understand dashboard and simple interface for viewing heart disease insights.
NFR-2	Security	Patient data should be protected and handled securely to ensure privacy and prevent unauthorized access.
NFR-3	Reliability	The system should consistently process and analyze health data without errors or crashes.
NFR-4	Performance	The system should analyze and display large datasets quickly with minimal delay.
NFR-5	Availability	The dashboard and analytics platform should be accessible whenever users need to view insights.

NFR No.	Non-Functional Requirement	Description
NFR-6	Scalability	The system should be able to handle increasing amounts of healthcare data in the future.

3.3 Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

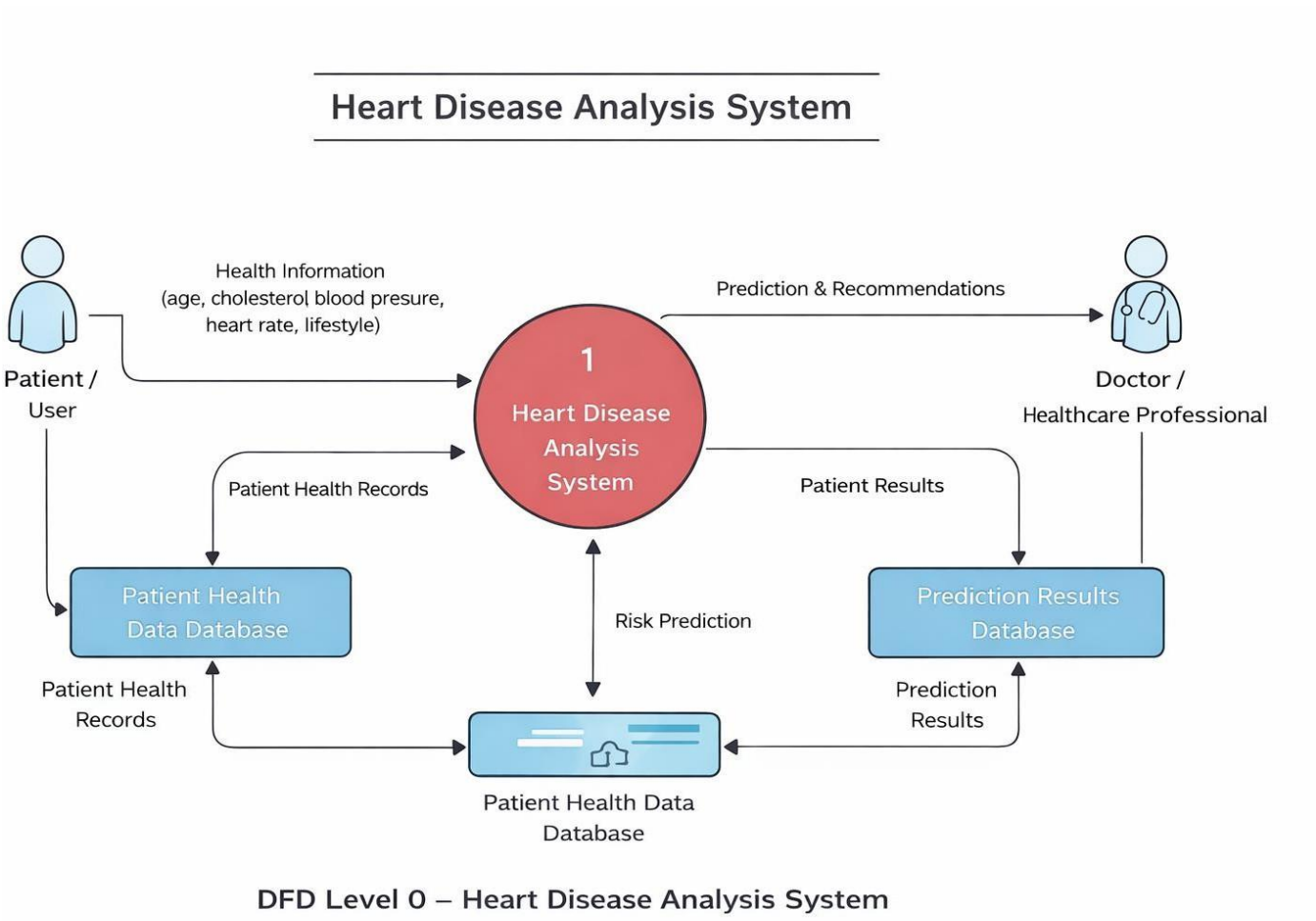


Figure 3 Data Flow Dig

Fig 1 Level 0 DFD Diagram

User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria
Patient / User	Registration	USN-1	As a user, I can register on the platform by entering my personal and health details.	User account is successfully created and accessible.
Patient / User	Login	USN-2	As a user, I can log into the system using my registered credentials.	User can securely access their dashboard.
Patient / User	Health Data Entry	USN-3	As a user, I can enter my medical parameters such as age, blood pressure, cholesterol, and heart rate.	System successfully stor the health data.
Patient / User	Heart Disease Prediction	USN-4	As a user, I can analyze my health data to check the risk of heart disease.	System generates a prediction result with ris level.
Patient / User	Dashboard	USN-5	As a user, I can view my health reports and prediction results on a dashboard.	Dashboard displays char and prediction outcome
Doctor / Healthcare Professional	Data Review	USN-6	As a doctor, I can review patient health data and prediction results.	Doctor can access patien reports and analysis.
Administrator	Data Management	USN-7	As an admin, I can manage patient records and system data.	Admin can update or delete records if necessary.
Administrator	System Monitoring	USN-8	As an admin, I can monitor system performance and data flow.	System logs and monitoring tools are available.

3.4 Technology Stack (Architecture & Stack)

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Architectural Diagram :-

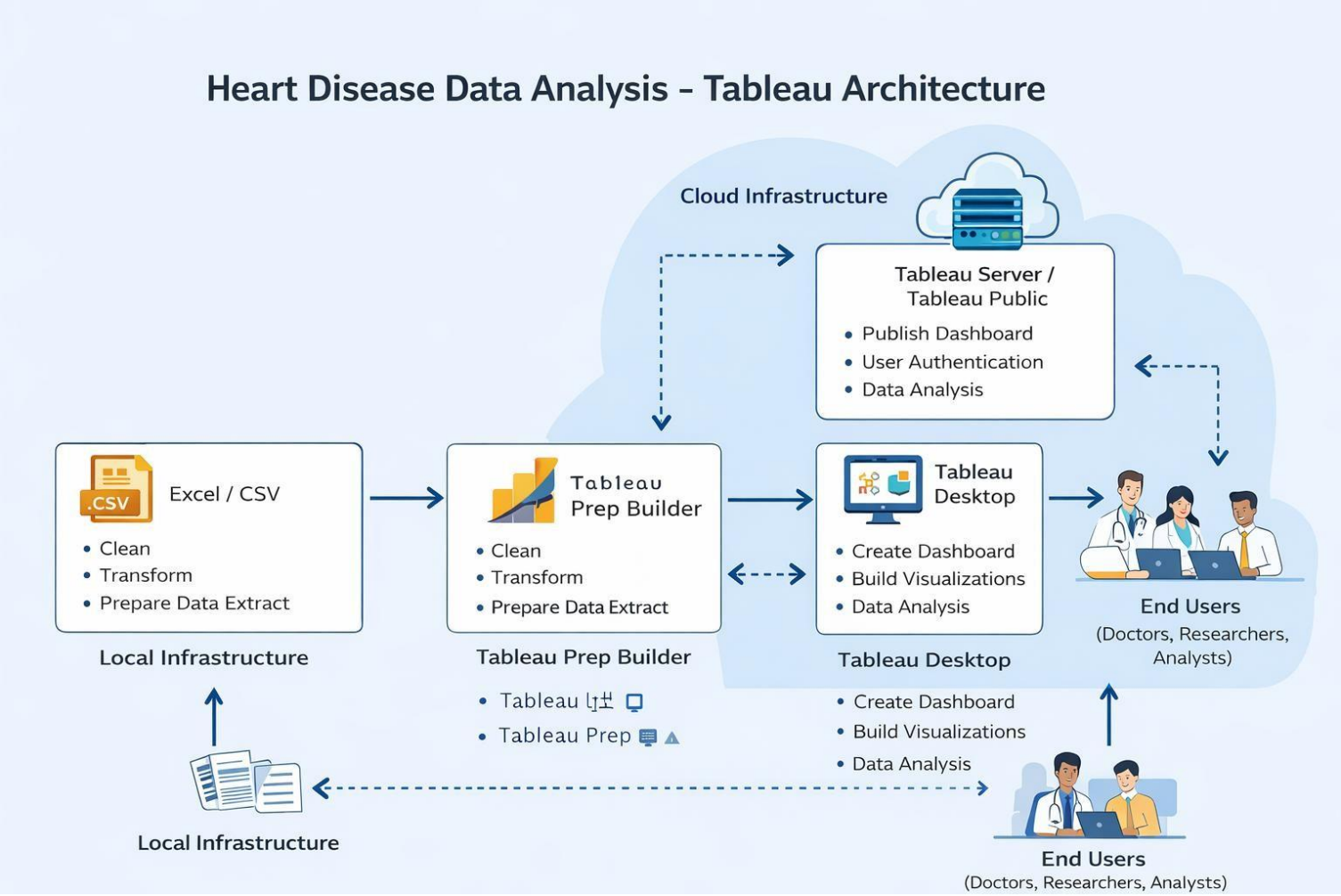


Figure 4 System Architecture

Table-1 : Components & Technologies:

S.No	Component	Description
1	User Interface	Interactive dashboards where users can explore heart disease data, filter patient attributes, and visualize risk factors through charts and graphs.
2	Application Logic-1	Data preparation process including cleaning, filtering, and structuring the heart disease dataset before visualization.
3	Application Logic-2	Data transformation operations such as joins, unions, calculated fields, and aggregation of patien data.
4	Application Logic-3	Data visualization and dashboard logic including filters, parameters, calculated fields, and interac charts.

S.No	Component	Description
5	Database	Source dataset containing patient health attributes like age, cholesterol, blood pressure, and hea disease indicators.
6	Cloud Database	Optional cloud storage used to host datasets for remote access and dashboard connectivity.
7	File Storage	Storage for datasets, Tableau workbook files (.twb / .twbx), and prepared data extracts.
8	External API-1	Data connection interface used to import datasets from external files.
9	External API-2	Integration used to publish and share dashboards online.
10	Data Analysis Model	Analytical visualizations used to identify patterns, correlations, and risk factors in heart disease d
11	Infrastructure (Server / Cloud)	Platform used to publish and share dashboards with users.

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Tools used for visualization and data preparation in the project.	Tableau Desktop, Tableau Prep Builder
2	Security Implementations	Secure access to dashboards and datasets through authentication and role-based access control.	Tableau Server Security, User Authentication
3	Scalable Architecture	Dashboards and data extracts can handle large datasets and multiple user access.	Tableau Server Architecture
4	Availability	Dashboards can be accessed anytime through Tableau Server or Tableau Public.	Tableau Cloud / Tableau Public
5	Performance	Use of Tableau extracts and optimized dashboards ensures faster loading and interactive performance.	Tableau Data Extracts (.hyper), Dashboard Optimization

4. PROJECT DESIGN

4.1 Solution Fit

Define CS, fit into CC

1. CUSTOMER SEGMENT (CS)

The primary customers for this solution include:

- Healthcare analysts and hospital management teams
- Doctors and medical researchers studying heart disease patterns
- Public health organizations monitoring cardiovascular risk factors
- Medical students and researchers analyzing healthcare datasets
- Government health departments tracking disease trends

These users require clear and interactive visualization tools to analyze heart disease data and identify risk patterns.

6. CUSTOMER CONSTRAINTS

Customers may face several constraints:

- Limited technical knowledge in data visualization tools
- Lack of time to analyze complex datasets manually
- Budget limitations for advanced analytics tools
- Difficulty integrating multiple healthcare datasets
- Limited access to real-time healthcare data

5. AVAILABLE SOLUTIONS

Current solutions customers use include:

- Excel spreadsheets for manual data analysis
- Basic statistical analysis tools
- Paper reports and traditional healthcare statistics
- Static charts and graphs

Limitations of these solutions:

- Limited visualization capabilities
- Difficult to analyze large datasets
- No interactive exploration of data
- Time-consuming analysis process

Explore AS, differentiate

Focus on J&P, tap into BE, understand RC

2. JOBS-TO-BE-DONE / PROBLEMS (J&P)

J&P

Customers need to:

- Analyze large datasets related to heart disease patients
- Identify major risk factors such as age, cholesterol, blood pressure, and lifestyle
- Detect patterns and correlations between health indicators and heart disease
- Monitor trends across different demographic groups
- Present healthcare insights visually for decision making

Current problems include:

- Raw healthcare data is complex and difficult to interpret
- Traditional spreadsheet analysis is time-consuming
- Lack of clear visual dashboards for medical insights
- Difficulty identifying risk patterns quickly

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9. PROBLEM ROOT CAUSE

RC

The root causes behind the problem include:

- Large volumes of healthcare data that are difficult to analyze manually
- Lack of effective visualization tools in traditional healthcare reporting
- Growing complexity of health indicators affecting heart disease
- Need for better decision-support tools in healthcare analysis
- Increasing cases of heart disease requiring deeper data insights

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7. BEHAVIOUR

BE

Customers currently attempt to solve the problem by:

- Collecting healthcare datasets from hospitals and research sources
- Using spreadsheets or statistical tools to analyze patient data
- Generating reports and charts for presentation
- Studying trends such as age groups, cholesterol levels, and blood pressure
- Sharing findings through presentations or research papers

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Focus on J&P, tap into BE, understand RC

3. TRIGGERS

TR

Customers are triggered to act when:

- Hospitals need data-driven insights for patient risk assessment
- Healthcare researchers need to analyze heart disease trends
- Governments require statistics on cardiovascular diseases
- Medical professionals want interactive dashboards instead of raw datasets
- Increasing heart disease cases require better analysis tools

4. EMOTIONS: BEFORE / AFTER

EM

Before using the solution:

- Confused by complex medical datasets
- Frustrated with manual analysis
- Difficulty identifying meaningful insights
- Lack of clarity in healthcare data interpretation

After using the solution:

- Confident in understanding heart disease patterns
- Better decision making using visual dashboards
- Faster analysis of medical data
- Improved ability to communicate insights

10. YOUR SOLUTION

SL

The proposed solution is a Tableau-based Heart Disease Analysis Dashboard that:

- Visualizes heart disease data using interactive dashboards
- Identifies risk factors such as age, cholesterol, blood pressure, and lifestyle
- Enables filtering and exploration of patient data
- Displays trends and correlations in an easy-to-understand format
- Supports healthcare professionals in data-driven decision making

The system uses **Tableau Desktop and Tableau Prep** to clean, analyze, and visualize healthcare datasets efficiently.

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8. CHANNELS of BEHAVIOUR

CH

8.1 ONLINE

- Healthcare research websites
- Medical data repositories (Kaggle, WHO datasets)
- Tableau dashboards and online visualization platforms
- Healthcare analytics forums and research communities

8.2 OFFLINE

- Hospitals and healthcare institutions
- Medical conferences and seminars
- Academic research institutions
- Government health departments

4.2 Proposed Solution

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	• Heart disease is a leading cause of death worldwide. • Large healthcare datasets are difficult to analyze manually. • Lack of clear visualization makes identifying risk factors challenging. • Need a system to analyze and visualize heart disease data effectively.
2.	Idea / Solution description	• Develop an interactive Heart Disease Analysis Dashboard using Tableau. • Use Tableau Prep Builder for data cleaning and preparation. • Visualize key health indicators like age, cholesterol, BP, and chest pain. • Enable easy analysis through filters and interactive dashboards.
3.	Novelty / Uniqueness	• Interactive visualization of complex healthcare datasets. • Real-time exploration of relationships between health parameters. • Simplifies medical data analysis through visual dashboards.
4.	Social Impact / Customer Satisfaction	• Helps identify heart disease risk patterns. • Supports better healthcare decision making. • Raises awareness about heart disease risk factors. • Improves data-driven healthcare analysis.
5.	Business Model (Revenue Model)	• Dashboard solutions for hospitals and healthcare organizations. • Subscription-based analytics dashboard services. • Data visualization consulting for healthcare institutions.
6.	Scalability of the Solution	• Can handle larger healthcare datasets. • Can integrate additional medical data sources. • Can be extended to analyze other diseases in the future.

4.3 Solution Architecture

The solution architecture for the Heart Disease Data Analysis project focuses on transforming raw healthcare data into meaningful visual insights using Tableau and Tableau Prep Builder. The system collects patient health records, prepares and cleans the data, and presents analytical dashboards that help identify heart disease risk patterns.

The architecture connects data sources, data preparation tools, and visualization platforms to ensure smooth data processing and reporting.

Architecture Overview

1. Data Source Layer

- Contains raw heart disease dataset.
- Includes patient attributes like age, cholesterol, blood pressure, heart rate, etc.
- Data stored in CSV or structured dataset format.

2. Data Preparation Layer (Tableau Prep Builder)

- Cleans and transforms raw dataset.
- Handles missing values and incorrect entries.
- Creates calculated fields and structured tables.
- Outputs a refined dataset for analysis.

3. Data Storage Layer

- Prepared dataset stored as Tableau Extract (.hyper) or structured dataset.
- Ensures fast query processing and efficient visualization.

4. Visualization Layer (Tableau Desktop)

- Connects to prepared dataset.
- Creates dashboards, charts, and analytical views.
- Enables exploration of heart disease risk indicators.

5. Reporting & Insights Layer

- Interactive dashboards show patterns and trends.
- Helps identify high-risk patient groups.
- Provides decision support for healthcare insights.

Data Flow in the System

1. Heart disease dataset is collected from healthcare records or public datasets.
2. The dataset is imported into Tableau Prep Builder for cleaning and transformation.
3. Processed data is exported as a structured dataset or Tableau extract.

4. Tableau Desktop connects to the processed data.
5. Interactive dashboards are created for visualization and analysis.
6. Insights are generated for identifying heart disease patterns and risk factors.

Solution Architecture Diagram:-

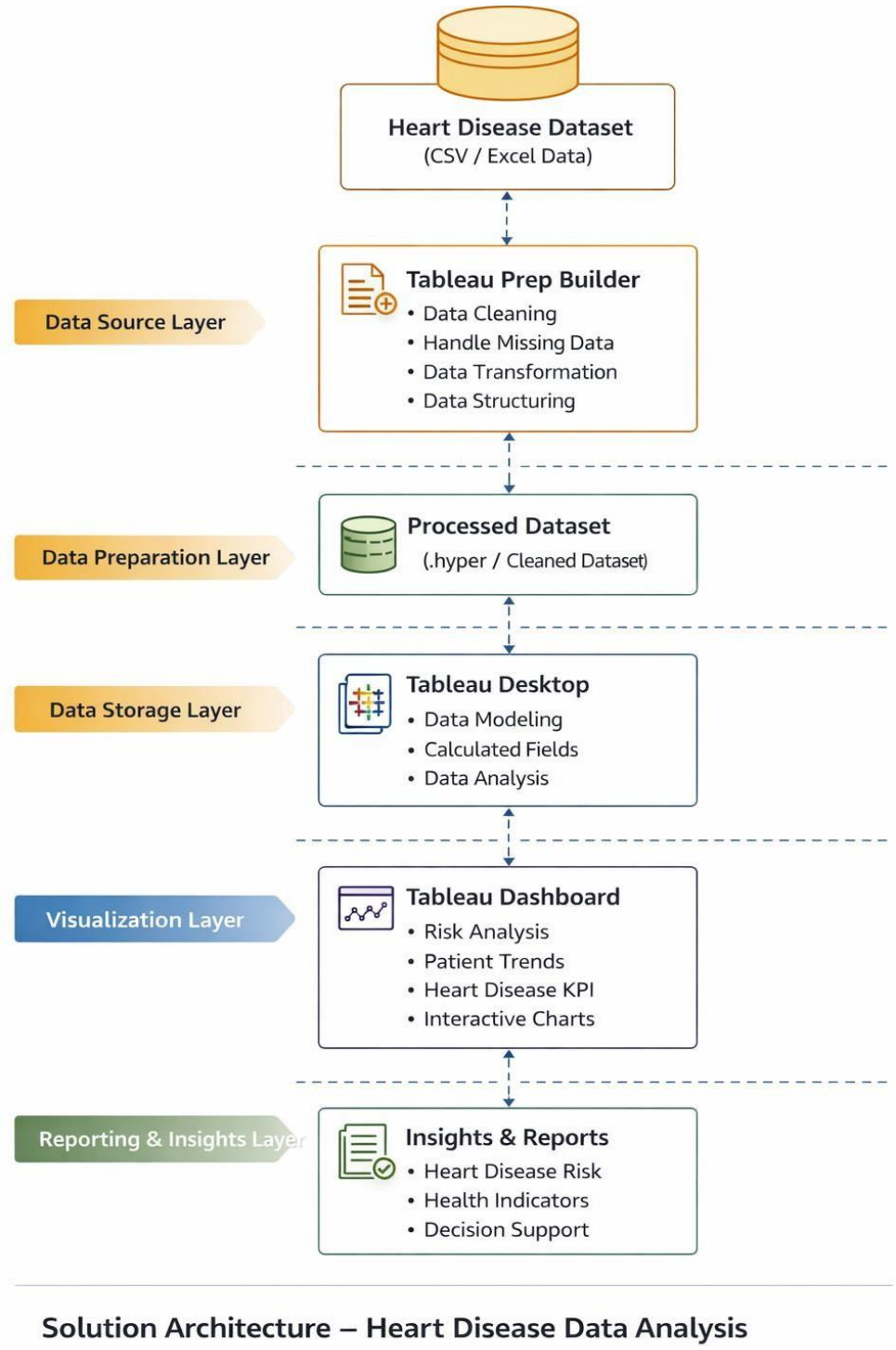


Figure 5 Solution Architecture

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Product Backlog, Sprint Schedule, and Estimation

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Data Collection	USN-1	As a developer, I can collect the heart disease dataset from public sources for analysis.	2	High	Team
Sprint-1	Data Preparation	USN-2	As a developer, I can import the dataset into Tableau Prep Builder for cleaning.	2	High	Team
Sprint-2	Data Cleaning	USN-3	As a developer, I can remove null values and incorrect records from the dataset.	2	High	Team
Sprint-2	Data Transformation	USN-4	As a developer, I can transform columns such as age, cholesterol, and blood pressure for analysis.	2	High	Team
Sprint-3	Data Visualization	USN-5	As a user, I can view a dashboard showing heart disease risk distribution by age and gender.	3	High	Team
Sprint-3	Dashboard Creation	USN-6	As a user, I can analyze heart disease cases using interactive charts in Tableau.	3	High	Team
Sprint-4	Risk Analysis	USN-7	As a user, I can view the relationship between cholesterol levels and heart disease risk.	2	Medium	Team
Sprint-4	Trend Analysis	USN-8	As a user, I can identify trends of heart disease cases across different age groups.	2	Medium	Team
Sprint-5	Interactive Filters	USN-9	As a user, I can filter dashboard data by age, gender, and chest pain type.	2	High	Team
Sprint-5	Dashboard Optimization	USN-10	As a developer, I can improve visualization clarity using colors and labels.	2	Medium	Team
Sprint-6	Testing	USN-11	As a developer, I can test dashboard accuracy and verify dataset calculations.	2	High	Team
Sprint-6	Deployment	USN-12	As a developer, I can publish the final dashboard on Tableau Public or Tableau Server.	2	High	Team

Project Tracker, Velocity & Burndown Chart:

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed	Sprint Release Date
Sprint-1	4	2 Days	25 Feb 2026	26 Feb 2026	4	26 Feb 2026
Sprint-2	4	2 Days	27 Feb 2026	28 Feb 2026	4	28 Feb 2026
Sprint-3	6	2 Days	01 Mar 2026	02 Mar 2026	6	02 Mar 2026
Sprint-4	4	2 Days	03 Mar 2026	04 Mar 2026	4	04 Mar 2026
Sprint-5	4	2 Days	05 Mar 2026	06 Mar 2026	4	06 Mar 2026
Sprint-6	4	2 Days	07 Mar 2026	08 Mar 2026	4	08 Mar 2026

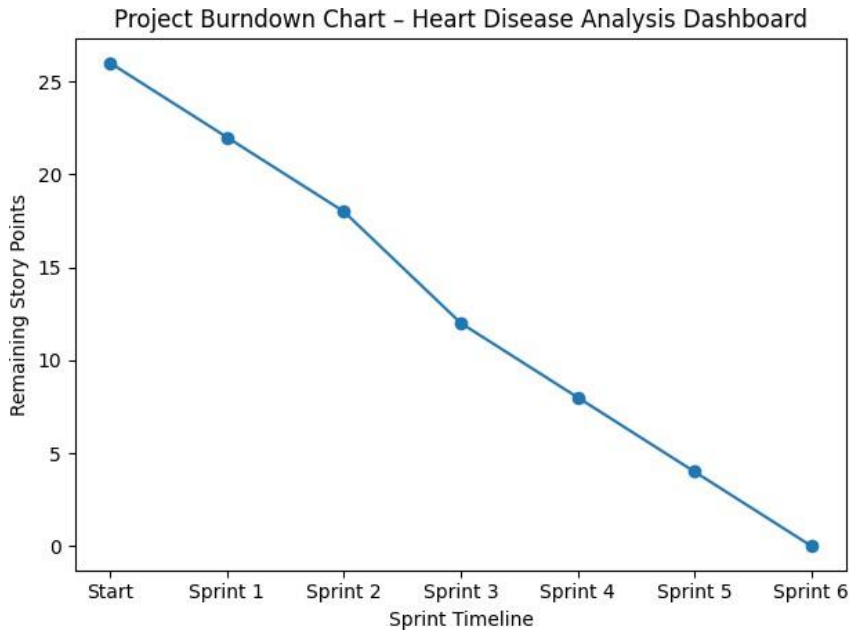
Velocity:
Imagine we have a 10-day sprint duration, and the velocity of the team is 20 (points per sprint). Let's calculate the team's average velocity (AV) per iteration unit (story points per day)

$$AV = \frac{sprint\ duration}{velocity} = \frac{20}{10} = 2$$

AV = Sprint Duration/Velocity = 20/10 = 2

Burndown Chart:

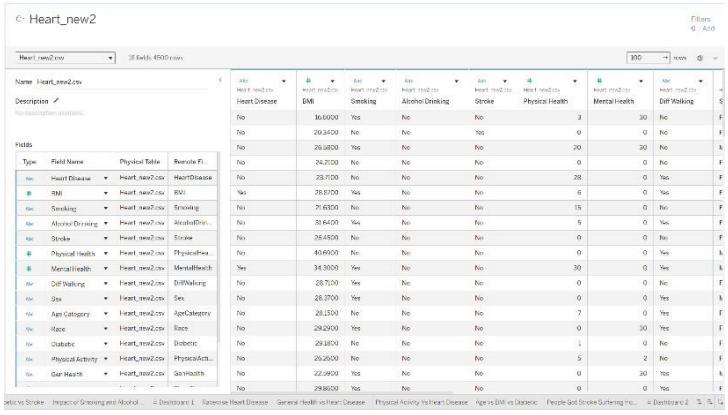
A burn down chart is a graphical representation of work left to do versus time. It is often used in agile software development methodologies such as Scrum. However, burn down charts can be applied to any project containing measurable progress over time.

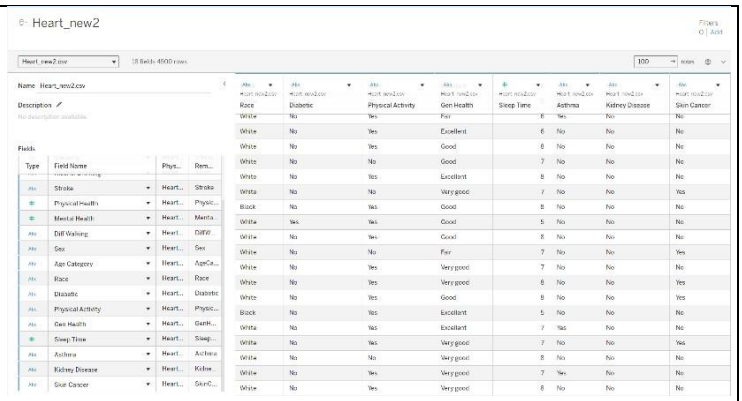


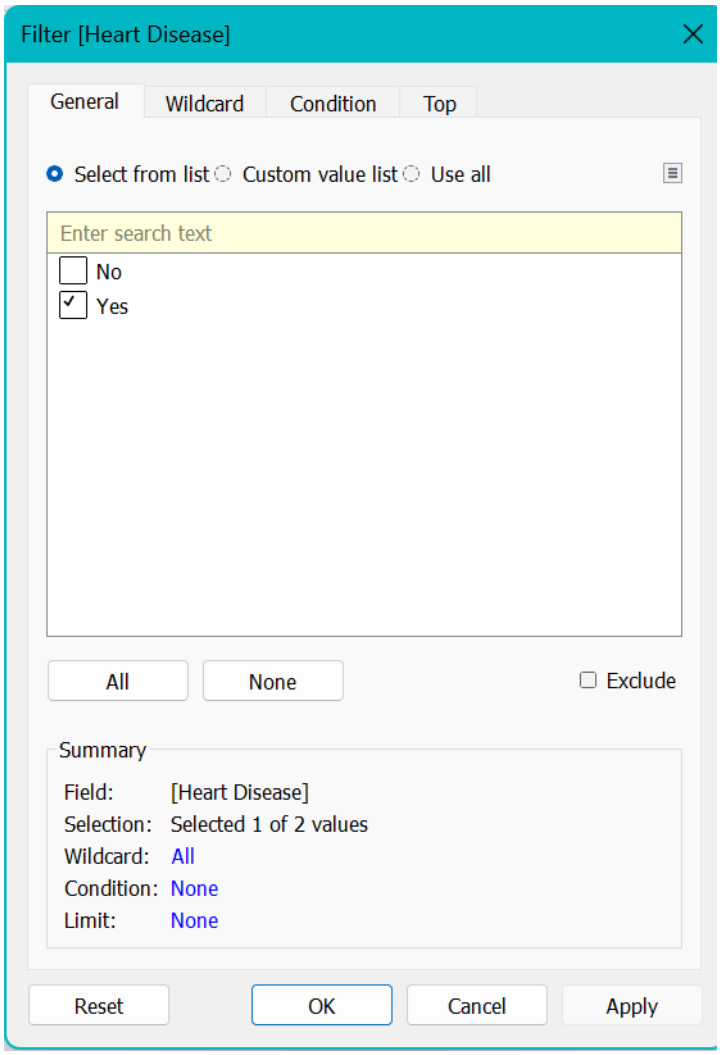
6. FUNCTIONAL AND PERFORMANCE TESTING

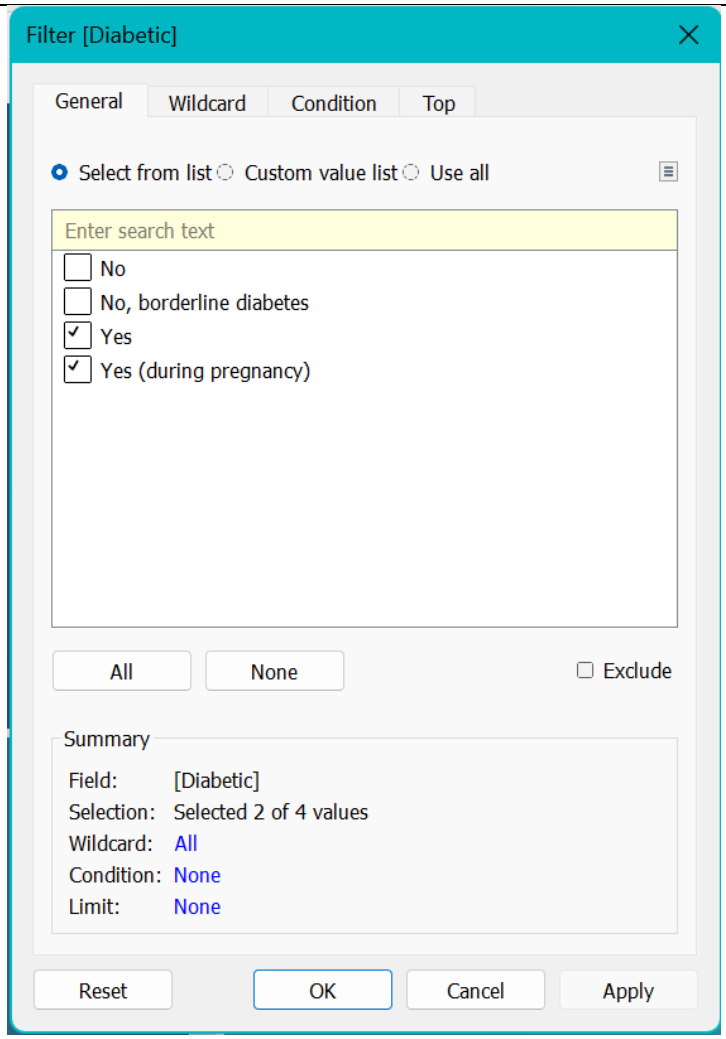
6.1 Model Performance Testing:

The performance testing phase evaluates how efficiently the Tableau dashboard processes, renders, and visualizes the dataset. The objective is to ensure that the dashboard provides fast, accurate, and interactive insights when users explore heart disease data.

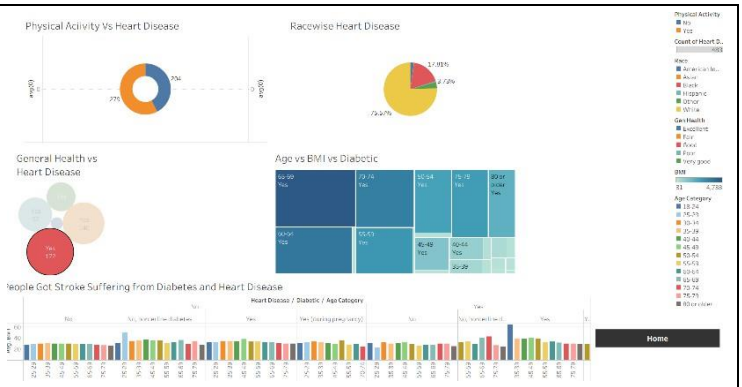
S.No.	Parameter	Screenshot / Values
1.	Data Rendered	**Description:** The heart disease dataset was successfully imported into Tableau from a CSV file. The dataset contains patient health indicators such as BMI, smoking habits, physical activity, diabetes status, age category, and other medical conditions. Tableau processed the dataset and generated visualizations efficiently. **Observations:** • Dataset Size: ~4500 records • Number of Attributes: 18 columns • Data Types: Dimensions and Measures • Rendering Time: Less than 2 seconds for visualization loading **Result:** The data was rendered smoothly in Tableau worksheets and dashboards without performance issues. Visualizations update dynamically based on filters and selections. **Screenshot:** 

		
2.	Data Preprocessing	**Description:** Data preprocessing was performed to ensure the dataset was clean and suitable for visualization. Tableau Prep Builder was used to structure and validate the data before loading it into Tableau Desktop. **Steps Performed:** • Removed duplicate records • Checked for missing or null values • Converted categorical fields into dimensions • Validated numerical measures such as BMI and SleepTime • Standardized column names for better readability **Result:** The cleaned dataset improved dashboard performance and ensured accurate visualizations.
3.	Utilization of Filters	**Description:** Interactive filters were added to the dashboard to allow users to analyze heart disease data based on different health factors. **Filters Implemented:** • Age Category • Gender • Smoking Habit • Physical Activity • Diabetes Status • General Health Condition **Result:** Users can dynamically explore the dataset and identify

		patterns affecting heart disease risk. Filters update visualizations instantly without delay. **Screenshot:** 

		
4.	Calculation fields Used	**Description:** Calculated fields were created to derive meaningful insights from the dataset and categorize patient health conditions. **Calculated Fields Created:** **1. BMI Category** Example Calculation: IF [BMI] < 18.5 THEN "Underweight" ELSEIF [BMI] >= 18.5 AND [BMI] < 25 THEN "Normal" ELSEIF [BMI] >= 25 AND [BMI] < 30 THEN "Overweight" ELSE "Obese" END **2. Health Risk Indicator** IF [HeartDisease] = "Yes" THEN "High Risk" ELSE "Low Risk" END

		Result: Calculated fields helped classify patients into different risk groups and improved the interpretability of visualizations.
5.	Dashboard design	**Description:** An interactive Tableau dashboard was designed to visualize patterns and trends related to heart disease. The dashboard integrates multiple charts to provide a comprehensive overview of patient health indicators. **Number of Visualizations / Graphs:** 6 **Visualizations Included:** 1. Heart Disease Distribution (Pie Chart) 2. Heart Disease by Age Category (Bar Chart) 3. Smoking vs Heart Disease (Stacked Bar Chart) 4. BMI Distribution (Histogram) 5. Physical Activity vs Heart Disease Risk (Bar Chart) 6. General Health Condition Analysis (Heatmap) **Result:** The dashboard provides a clear and interactive interface for analysing factors associated with heart disease risk. **Screenshot:**

		
6	Story Design	**Description:** A Tableau Story was created to present analytical insights in a step-by-step format. This helps stakeholders understand patterns and key findings from the dataset. **Number of Visualizations / Graphs:** 5 **Story Flow:** 1. Introduction to Heart Disease Dataset 2. Demographic Analysis of Patients 3. Lifestyle Factors Affecting Heart Disease 4. Health Indicators and Risk Levels 5. Key Insights and Observations **Result:** The Tableau story effectively communicates analytical insights and supports decision-making through visual storytelling. **Screenshot:**

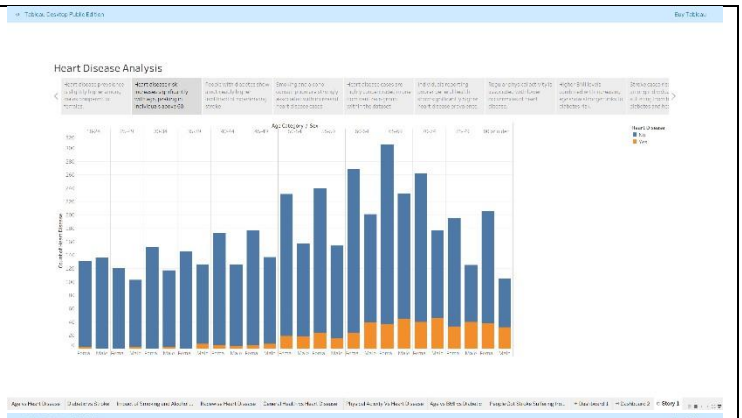


Tableau Desktop Public Edition
Buy Tableau
Heart Disease Analysis
Age Category / Sex
Heart Disease
No
Yes

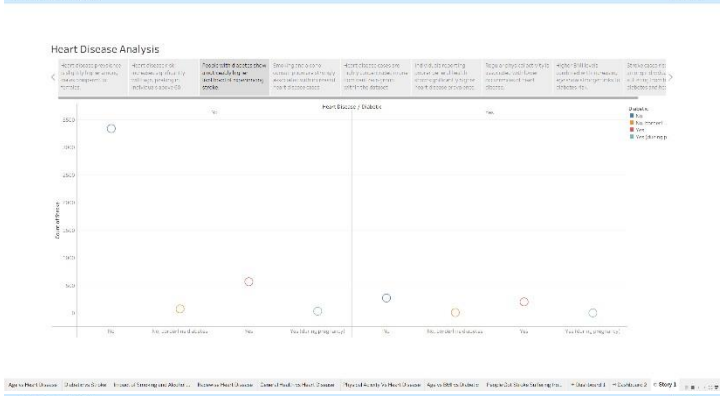


Tableau Desktop Public Edition
Buy Tableau
Heart Disease Analysis
Heart Disease / Diabetes
Diabetes
No
No, borderline diabetes
Yes
Yes (during preg.)

Tableau Desktop Public Edition
Buy Tableau
Heart Disease Analysis
Smoking Alcohol
Stroke
No
Yes
Frequent Heavy Drinker

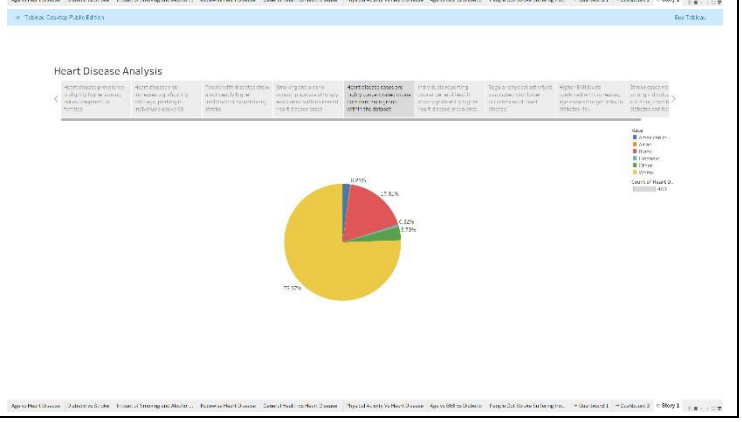
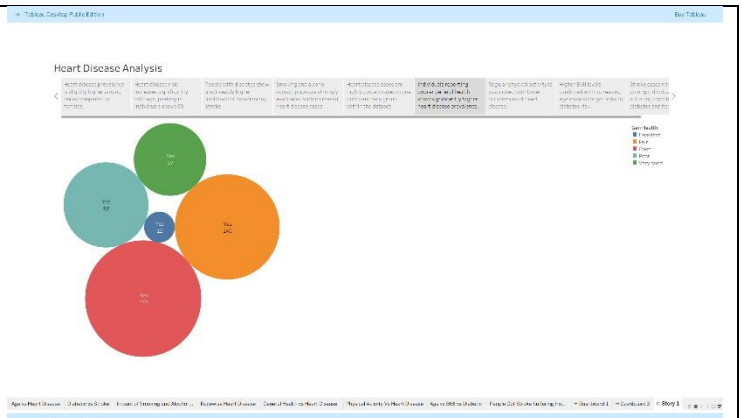
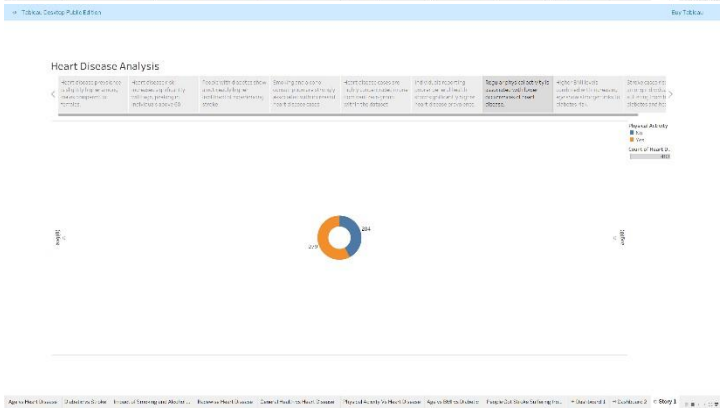


Tableau Desktop Public Edition
Buy Tableau
Heart Disease Analysis
Value
Asian
Black
Hispanic
Other
White
Count of Heart D...
410
0.91%
17.8%
0.62%
2.73%
78.37%


Heart Disease Analysis


Heart Disease Analysis

Heart Disease Analysis

Heart Disease Analysis

7. Results

Figure 6 Dashboard Page 1

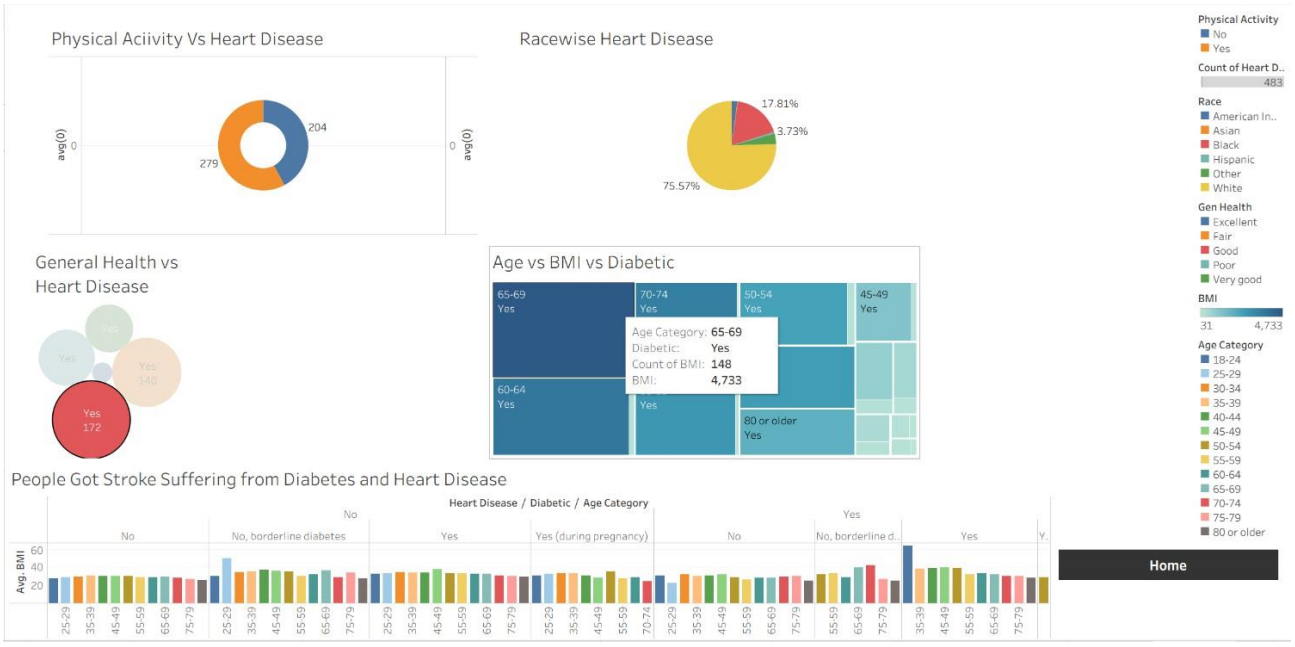


Figure 7 Dashboard Page 2

8. ADVANTAGES & DISADVANTAGES

Advantages

1. Early Risk Identification
   The analysis helps identify patterns and indicators associated with heart disease, enabling early risk detection.
2. Data-Driven Decision Making
   Healthcare professionals can use insights from the dashboard to make informed decisions based on data rather than assumptions.
3. Interactive Visualization
   Using Tableau dashboards, users can explore data interactively through filters, charts, and drill-down capabilities.
4. Improved Healthcare Insights
   Visual analytics helps understand relationships between factors such as age, cholesterol level, blood pressure, and heart disease risk.
5. Efficient Data Exploration
   Large datasets can be analyzed quickly using Tableau visualizations, making it easier to identify trends and anomalies.
6. User-Friendly Interface
   Tableau provides intuitive dashboards that can be easily understood by healthcare professionals and stakeholders without technical expertise.
7. Better Reporting and Presentation
   Visual dashboards make it easier to present findings to stakeholders, hospitals, and researchers.

Disadvantages

1. Limited to Available Data
   The insights are dependent on the dataset used. If the data is incomplete or outdated, the analysis may not be fully accurate.
2. No Direct Medical Diagnosis
   The project only provides analytical insights and cannot replace professional medical diagnosis or clinical evaluation.
3. Data Quality Dependency
   Errors or inconsistencies in the dataset can affect visualization accuracy and insights.
4. Limited Predictive Capability
   Since the project focuses mainly on data visualization, it may not include advanced predictive modeling.
5. Requires Tableau Knowledge
   Users need basic knowledge of Tableau to modify dashboards or perform advanced analysis.
6. Privacy and Data Sensitivity
   Healthcare data is sensitive, and improper handling can raise privacy concerns.

9. CONCLUSION

The Heart Disease Analysis Project uses data visualization techniques to analyze healthcare data and identify patterns related to heart disease risk factors. By utilizing Tableau and Tableau Prep Builder, the project transforms raw healthcare data into meaningful and interactive visual dashboards that help users easily understand complex information.

Through various visualizations, the project highlights key relationships between factors such as age, cholesterol level, blood pressure, chest pain type, and heart disease occurrence. These insights can assist healthcare professionals and analysts in understanding trends and identifying high-risk groups.

Overall, the project demonstrates how data analytics and visualization tools can support better decision-making in the healthcare sector. Although it does not replace medical diagnosis, it provides valuable analytical insights that can contribute to improved awareness, early risk identification, and data-driven healthcare analysis.

10. FUTURE SCOPE

- **Integration with Real-Time Healthcare Data**
  The project can be enhanced by integrating real-time hospital or healthcare data to provide more up-to-date insights and monitoring.
- **Advanced Predictive Analytics**
  Machine learning models can be incorporated to predict the likelihood of heart disease more accurately based on patient health parameters.
- **Integration with Healthcare Systems**
  The dashboard can be connected with hospital management systems or electronic health records (EHR) to support better clinical decision-making.
- **Enhanced Interactive Dashboards**
  Additional visualizations and advanced filtering options can be added to improve user interaction and deeper data exploration.
- **Mobile and Web Accessibility**
  The dashboard can be deployed on web and mobile platforms so healthcare professionals can access insights anytime and anywhere.
- **Personalized Health Risk Monitoring**
  Future versions of the project could provide personalized risk analysis for patients based on their medical data.
- **Expansion with Larger Datasets**
  Incorporating larger and more diverse healthcare datasets can improve the reliability and usefulness of the analysis.

11. APPENDIX

Dataset Link :- https://github.com/Venki01a/Heart-Disease-Analysis/blob/main/Heart_new2.csv

GitHub:- https://github.com/Venki01a/Heart-Disease-Analysis.git

Project Demo Link:- https://drive.google.com/file/d/1w1sf8PR2xz6xfMq7-PijTCnwwIEX_dFt/view